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Porous g-C₃N₄ with nitrogen defects and cyano groups for excellent photocatalytic nitrogen fixation without co-catalysts

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ABSTRACT: The porous g-C₃N₄ with nitrogen defects and cyano groups (NC-g-C₃N₄) is prepared via an alkali-assisted heat treatment of urea. Alkali can break hydrogen bonds, which accelerate thermal polymerization of urea and formation of nitrogen defects/cyano groups. The presence of nitrogen defects extends the absorption of visible light to longer wavelengths region. The cyano groups can trap g-C₃N₄'s photoinduced electrons and therefore suppress charge recombination. The formation of porous structure increases the surface area and exposes more active sites. As a consequence, compared to pure g-C₃N₄, NC-g-C₃N₄ shows boosted visible photocatalytic nitrogen fixation activity (1.59 mmol h⁻¹g⁻¹) without co-catalysts.

Keywords: g-C₃N₄; nitrogen vacancies; cyano groups; porous structure; photocatalytic nitrogen fixation

1. Introduction

In the synthesis of nucleotides, amino acids, and other major biological compounds, nitrogen is one of the starting building blocks [1, 2]. However, nitrogen molecules are solidified by the strong N≡N triple bonds, which make molecular nitrogen unobtainable [3, 4]. The artificial nitrogen fixation has attracted worldwide

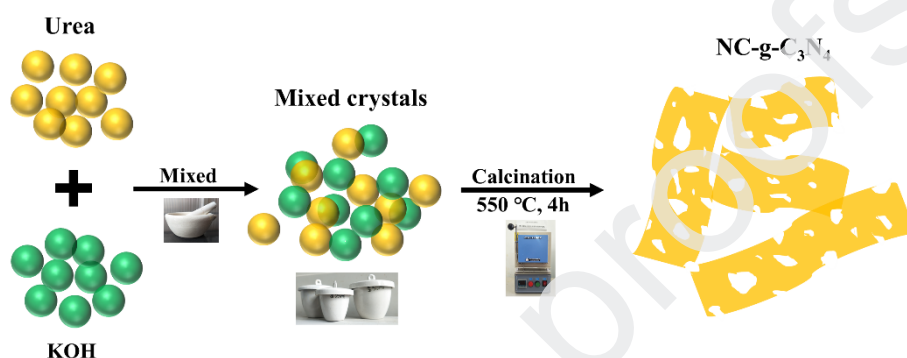
attention [5]. Since 1917, Haber-Bosch method became the most efficient nitrogen fixation artificial method [6]. Yet, it works under harsh conditions (673-823 K and 150-250 bar) [7-9]. Recently, Photocatalytic nitrogen fixation has aroused great interest due to the energy-saving and environment-friendly process [10]. Under light irradiation, photocatalysts absorb solar light and produce carriers, then the electrons could reduce adsorbed nitrogen into ammonia [11]. However, the photocatalytic efficiency is relatively low chiefly owing to the ineffective electron transfer [12].

To enhance the electron transfer property, the photocatalyst should possess more active sites [13], which can not only chemisorb N_2 molecules, but also establishes an efficient channel for photoexcited carriers transferring from photocatalyst to adsorbed N_2 [14]. So many kinds of photocatalysts with abundant active sites, such as nitrogen defective g- C_3N_4 , have been explored [15], which serves as the bridge for transferring electrons to N_2 . Different approaches, such as high temperature thermal polymerization [16], hydrothermal method [17], and reduction route [18], have been used for the preparation of g- C_3N_4 with nitrogen vacancies. Nevertheless, the above methods have limitations to accurately control the abundance and type of introduced nitrogen vacancies. Therefore, it is necessary to find out a simple method for *in-situ* introducing nitrogen vacancies into g- C_3N_4 during its synthesis [19].

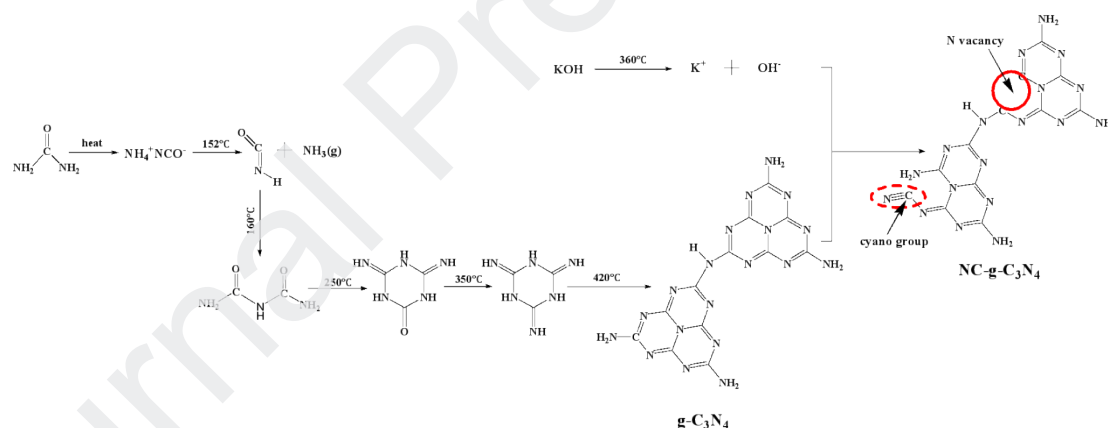
The preparation of g- C_3N_4 with abundant nitrogen vacancies and cyano groups (NC-g- C_3N_4) is regulated by the addition of KOH during the thermal condensation. Adding KOH preserves g- C_3N_4 's nanosheet structure and changes its chemical composition and porosity. The formation of nitrogen defects and cyano groups in the

g-C₃N₄ with porous feature and enlarged specific surface area can effectively narrow the bandgap, enhance the light absorption, and enable the separation of photo-generated carriers. Therefore, the NC-g-C₃N₄ can greatly enhance the visible photocatalytic nitrogen fixation activity.

2. Results and discussion



Scheme 1. Schematic diagram of the preparation of NC-g-C₃N₄.



Scheme 2. Schematic diagram for the formation of NC-g-C₃N₄.

The pure g-C₃N₄ is synthesized via direct hydrolysis of urea without KOH. As shown in Scheme 1, the NC-g-C₃N₄ is prepared by the hydrolysis of urea with adding KOH. The formation of NC-g-C₃N₄ was illustrated as Scheme 2. During the heating process, urea is initially decomposed to form ammonium cyanate (NH₄⁺NCO⁻). When the temperature reaches 152 °C, the urea completely decomposes to form cyanic acid

and ammonia. Upon continued heating to 160 °C, the reaction produces a biuret which is then reacted with cyanic acid to form cyanuric acid, after which ammelide formed. Further increasing temperature, ammeline is formed by ammelide reacting with NH_3 , and ammelide is further reacted to form melamine. Lastly, melamine is copolymerized, and abundant $\text{g-C}_3\text{N}_4$ is synthesized in the porcelain crucible [20]. It is worth noting that KOH will melt at 360 °C to produce OH^- , and then react with the intermediate amine groups in the thermal polymerization process to produce cyano groups ($-\text{C}\equiv\text{N}$) and nitrogen vacancies to form NC- $\text{g-C}_3\text{N}_4$ [21]. This strategy achieved the result of “one stone and two birds”.

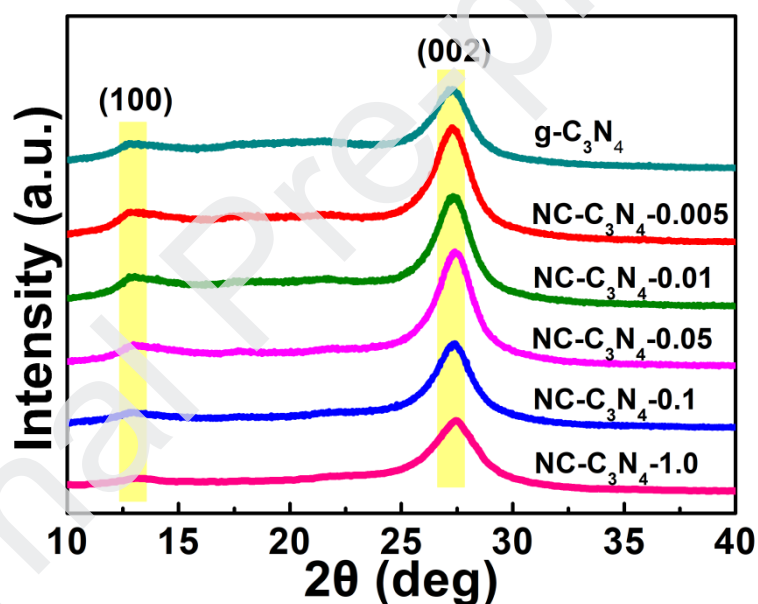


Fig. 1. XRD patterns of $\text{g-C}_3\text{N}_4$ and NC- $\text{g-C}_3\text{N}_4$ with different KOH usage.

For the pure $\text{g-C}_3\text{N}_4$ (Fig. 1), the peak located at 27.4° is ascribed to (002) plane of layered structure of $\text{g-C}_3\text{N}_4$ (JCPDS 87-1526) [22]. The other small peak at 13.0° is ascribed to (100) plane of tri-*s*-triazine [23]. The XRD patterns of NC- $\text{g-C}_3\text{N}_4$ are consistent with pure $\text{g-C}_3\text{N}_4$, which confirm the general structure of $\text{g-C}_3\text{N}_4$ after KOH treatment are well preserved. The (100) peak at 13.0° of NC- $\text{g-C}_3\text{N}_4$ samples becomes

weaker and broader than pure g-C₃N₄, corresponding with the decrease of in-plane repeating units [24]. This finding suggests that abundant vacancies exist in the NC-g-C₃N₄ framework. Besides, with increasing KOH usage, the (002) peak at 27.4° of NC-g-C₃N₄ shifts to higher 2 θ angles, indicate a decrease in the interlayer stacking distance [25]. Therefore, the direct condensation of the KOH pretreated urea certainly induce the partial loss of N atoms, thus the distance between interlayer stacking of g-C₃N₄ decreases.

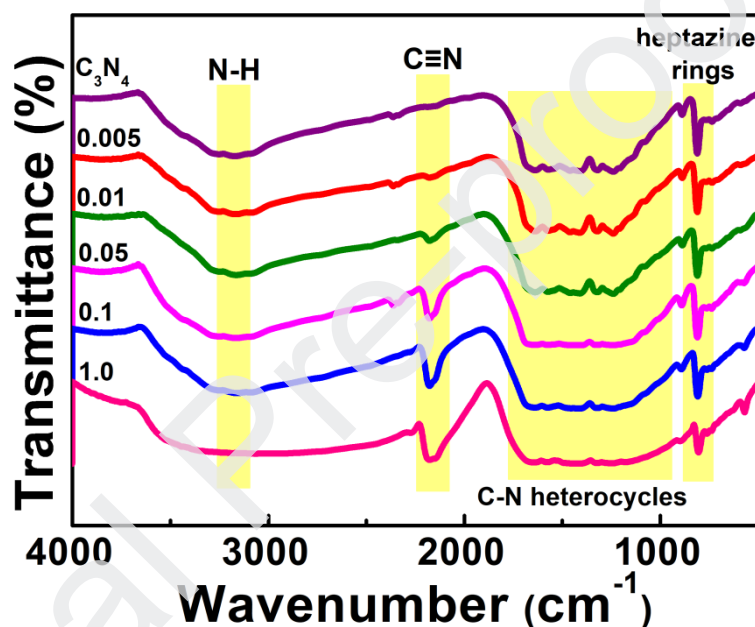


Fig. 2. FT-IR spectra of g-C₃N₄ and NC-g-C₃N₄ with different KOH usage.

As shown in Fig. 2, all samples have very similar FT-IR spectra, which indicates that the main molecular structure of g-C₃N₄ is mostly unaltered throughout the KOH-assisted process, in spite of the presence of nitrogen vacancies and cyano groups. The FT-IR spectrum of pure g-C₃N₄ (curve purple) shows that the absorption band at 812 cm⁻¹ is ascribed to out-of-plane bending mode of heptazine rings [26]. Several strong bands at 900-1800 cm⁻¹ range are ascribed to aromatic CN heterocycles stretching vibrations, while some wide peaks at around 3000-3300 cm⁻¹ are ascribed to stretching

vibrations of N-H [27]. For the NC-g-C₃N₄, with increasing KOH usage, N-H tensile peak intensity gradually decreases in the range of 3000-3300 cm⁻¹, while forming a new peak (2177 cm⁻¹), which corresponds to cyano groups (-C≡N) asymmetric stretching vibration. As displayed in Scheme 2, urea is reversibly isomerized into NH₄⁺CNO⁻ under heating treatment. After adding KOH, the OH⁻ could react with the amine group of urea-derived intermediate during pyrolysis to produce a cyano group [13, 16]. Besides, the intensities of peaks at 812 cm⁻¹ and 900-1800 cm⁻¹ decrease after precursor urea pretreated with KOH, suggesting the existence of nitrogen vacancies in NC-g-C₃N₄ [25]. The FT-IR spectroscopy results show that the addition of KOH reduces the proportion of N-H groups and induces the formation of cyano groups and nitrogen vacancies in NC-g-C₃N₄. The cyano groups are regarded as strong electron acceptors, because the electron pairs in cyano groups are strongly biased to N [28]. The formation of nitrogen defects and cyano groups could improve the photocatalytic performance.

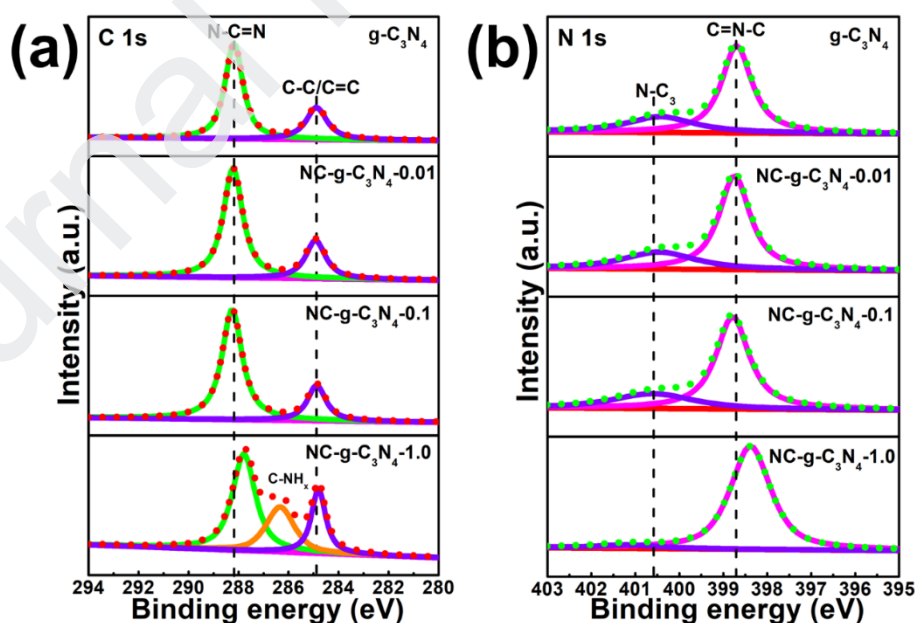


Fig. 3. (a) C 1s and (b) N 1s XPS spectra of g-C₃N₄ and NC-g-C₃N₄ with different KOH usage.

As shown in XPS survey spectra (Fig. S1), except for the sharp N 1s and C 1s peaks, the very small O 1s peak at 531.8 eV is observed, owing to surface adsorbed adventitious oxygen-containing species [29, 30]. Besides, the K 2s peak presents that K appears in the NC-g-C₃N₄-1.0 (Fig. S1). But K was lacked in NC-g-C₃N₄-0.01 and NC-g-C₃N₄-0.1, suggesting that the loading of K is determined by the concentration of KOH and K can not be decorated on g-C₃N₄ in the low concentration of solution. For pure g-C₃N₄ (Fig. 3a), two single C 1s peaks locating at 288.2 and 284.8 eV are assigned to heterocycle (N-C=N) and graphitic carbon (C=C and C-C) [31]. The C 1s peaks of NC-g-C₃N₄-0.01 and NC-g-C₃N₄-0.1 are consistent with that of g-C₃N₄. However, a new weak peak at 286.4 eV exists in NC-g-C₃N₄-1.0 (Fig. 3a), attributing to C-NH_x (x = 1, 2) of cyano groups [21]. The g-C₃N₄ and NC-g-C₃N₄ (Fig. 3b) with different KOH usage present two peaks at 398.6 and 400.7 eV, ascribing to tertiary carbon C=N-C (N₂C) and N-C₃ (N₃C) groups, respectively. Remarkably, the peak area ratio of N₃C/N₂C decreases with increasing KOH usage, showing that the tertiary N atoms are lost in the reaction [32]. The N/C atomic ratios were calculated as 1.38, 1.36, 1.20, and 0.87 for g-C₃N₄ and NC-g-C₃N₄-(0.01, 0.1, 1.0), respectively (Table S1). Compared with g-C₃N₄, the decrease of N/C atomic values in NC-g-C₃N₄ indicates the loss of lattice N and presence of N vacancies in NC-g-C₃N₄ [21].

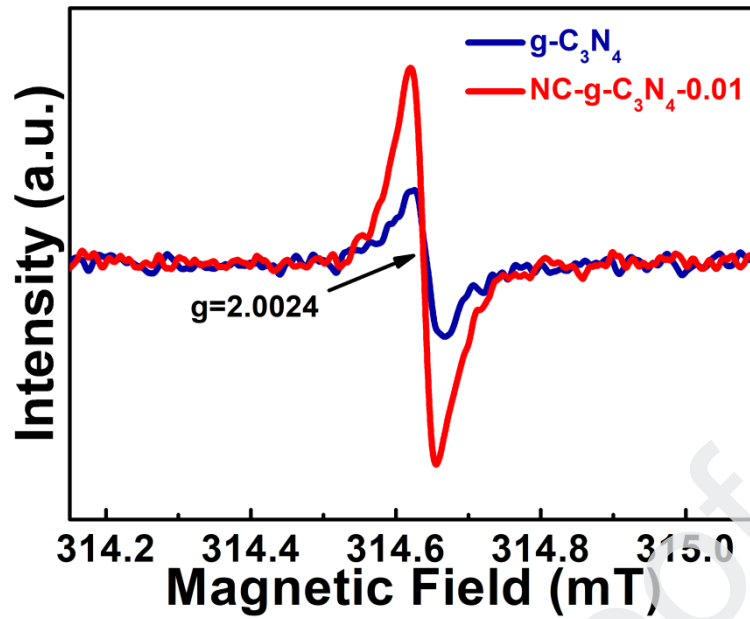


Fig. 4. Room-temperature EPR spectra of $g\text{-C}_3\text{N}_4$ and $\text{NC-}g\text{-C}_3\text{N}_4\text{-0.01}$.

Room temperature electron paramagnetic resonance (EPR) spectroscopy further confirms the formation of N vacancies in $\text{NC-}g\text{-C}_3\text{N}_4\text{-0.01}$ [15]. As shown in Fig. 4, a strong symmetrical signal locating at the g factor of 2.0024 was detected in both $g\text{-C}_3\text{N}_4$ and $\text{NC-}g\text{-C}_3\text{N}_4\text{-0.01}$, and all exhibited a similar single Lorentzian line. Moreover, compared with $g\text{-C}_3\text{N}_4$, the signal intensity of $\text{NC-}g\text{-C}_3\text{N}_4\text{-0.01}$ is considerably enhanced, owing to N defects in $\text{NC-}g\text{-C}_3\text{N}_4\text{-0.01}$, which increase the number of unpaired electrons [32]. Therefore, the enhancement of the EPR signal in $\text{NC-}g\text{-C}_3\text{N}_4\text{-0.01}$ further confirms the generation of nitrogen defects in $\text{NC-}g\text{-C}_3\text{N}_4\text{-0.01}$.

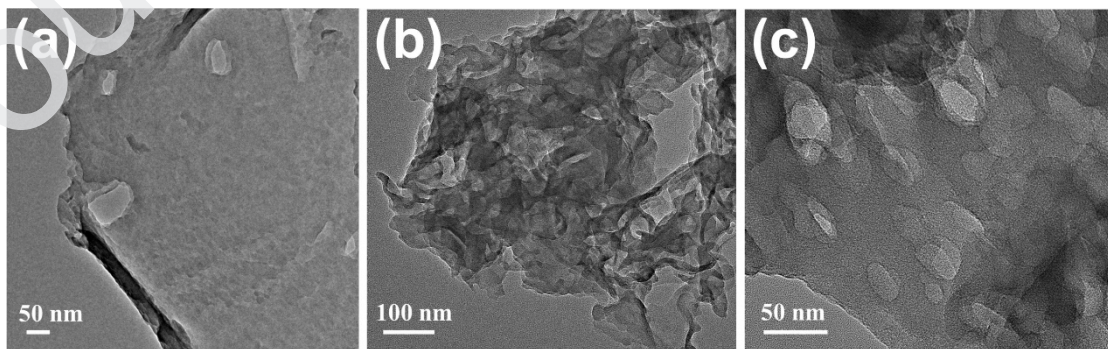


Fig. 5. TEM images of (a) $g\text{-C}_3\text{N}_4$ and (b, c) $\text{NC-}g\text{-C}_3\text{N}_4\text{-0.01}$.

To measure the morphology of NC-g-C₃N₄, TEM has been introduced (Fig. 5 and S2). The g-C₃N₄ (Fig. 5a) shows a sheet-like structure with the smooth surface [33]. As compared with a smooth surface of g-C₃N₄, abundant wrinkles and pores appear in NC-g-C₃N₄ (Fig. S2, 5b and 5c), due that, during thermal polymerization process, KOH etches the melon unit of g-C₃N₄ in the edge, and forms fragment or pore structure. The porous morphology for the prepared NC-g-C₃N₄ materials might provide larger surface area for accommodating active sites [34]. Moreover, the morphology of NC-g-C₃N₄ is also investigated by SEM. Dense and thick layers are detected in g-C₃N₄ (Fig. S3a). From Fig. S3b-f, the SEM image shows that NC-g-C₃N₄ possesses abundant pores, except for g-C₃N₄-1.0, which shows a large particle construction, due that the morphology can be broken at a high KOH dosage. Therefore, we can know that the addition of KOH modifies the chemical composition and mesoporosity of g-C₃N₄, but retains sheet structure [19].

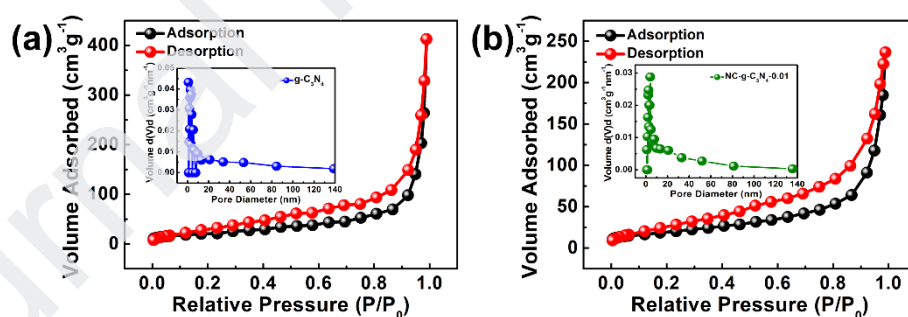


Fig. 6. N₂ adsorption-desorption isotherms and corresponding pore size distribution curves (insert) of (a) g-C₃N₄ and (b) NC-g-C₃N₄-0.01.

The specific surface areas of NC-g-C₃N₄ with different KOH usage were measured by nitrogen adsorption-desorption (Fig. 6 and S4). All of the samples present a characteristic type-IV isotherm. NC-g-C₃N₄-0.005 (86.6 m² g⁻¹) and NC-g-C₃N₄-0.01

(70.1 m² g⁻¹) show larger BET surface area than g-C₃N₄ (40.9 m² g⁻¹) (Table S2). Adding KOH create abundant pores with larger specific surface area. However, when the mass of KOH is increased to 0.05, 0.1, and 1.0 g, the pore volume and BET surface area of NC-g-C₃N₄-0.05, NC-g-C₃N₄-0.1 and NC-g-C₃N₄-1.0 are sharply decreased (Table S2). This is because the excessive KOH induced collapse of the nanosheet structure. The TEM images of NC-g-C₃N₄-1.0 (Fig. S5) shows that when the added amount of KOH is 1 g, the original nanosheet structure of g-C₃N₄ is partially destroyed, resulting in a greatly reduced surface area. Interestingly, the NC-g-C₃N₄-0.01 present largest pore diameter (3.985 nm), which facilitates the absorption and diffusion of reactants.

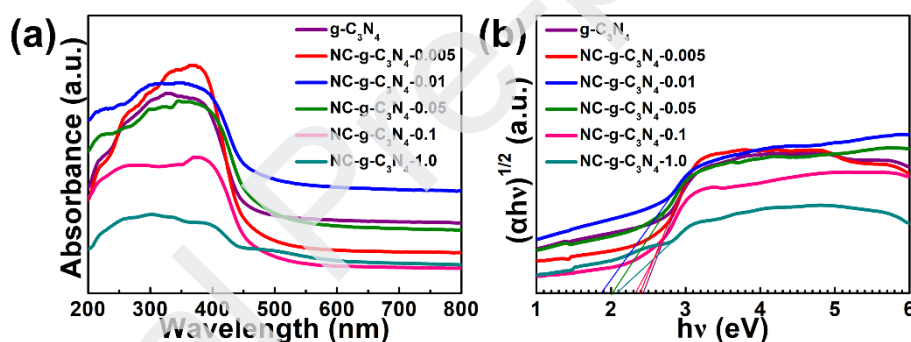


Fig. 7. (a) DRS and (b) band gap values of g-C₃N₄ and NC-g-C₃N₄ with different KOH usage.

The g-C₃N₄ presents the absorption edge at 455 nm (Fig. 7a) [35]. However, with the increasing amount of KOH, the absorption of NC-g-C₃N₄ in the visible regions increase, and the absorption of UV light is weakened, consistent with the primary color variance of NC-g-C₃N₄ from weak yellow to dark yellow (Fig. S6). It is important that when the amount of KOH added is 0.01 g, the absorption intensity of ultraviolet light is minimally affected (curve red, Fig. 7a). Besides, the enhanced absorption of NC-g-

C_3N_4 probably also results from porous structure through the multiple scattering effects [36]. Fig. 7b shows calculated bandgaps (E_g) for $g-C_3N_4$ and NC- $g-C_3N_4$. The calculated E_g of $g-C_3N_4$ and NC- $g-C_3N_4$ -(0.005, 0.01, 0.05, 0.1, 1.0) are estimated to be 2.49, 2.45, 2.40, 2.38, 2.13 and 2.09 eV, respectively. A narrower band gap is achieved with increasing KOH usage, which should be related to the nitrogen vacancies [19], that are produced during the KOH-assisted process [37].

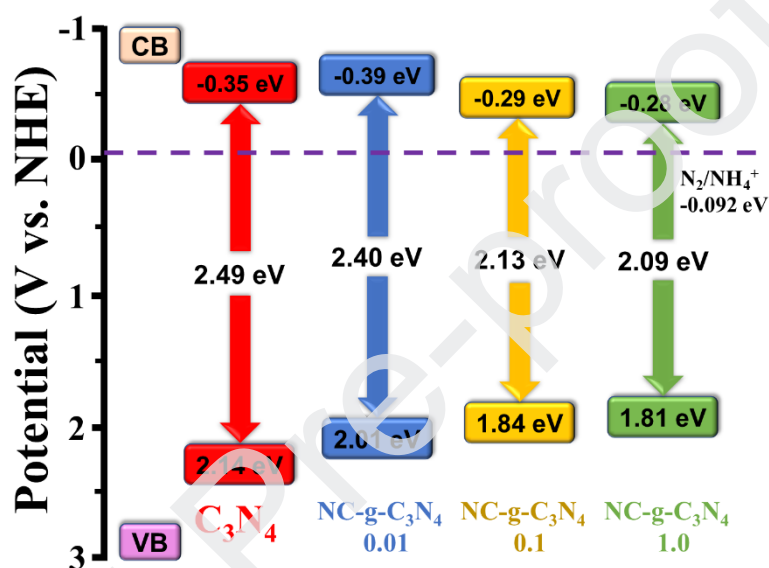


Fig. 8. The band structure alignments for $g-C_3N_4$ and NC- $g-C_3N_4$ (0.01, 0.1 and 1.0 g KOH).

Generally, the photocatalytic nitrogen fixation could occur mainly due to the position of valence band (VB) and conduction band (CB) in semiconductor. Therefore, the reduction potential of N_2 should be in the middle of CB and VB of the semiconductor [38]. The conduction band potentials (E_{CB}) of $g-C_3N_4$ and NC- $g-C_3N_4$ are evaluated by the Mott-Schottky curves. As shown in Fig. S7, the slopes of the Mott-Schottky curves are positive, indicating that both $g-C_3N_4$ and NC- $g-C_3N_4$ are n-type semiconductors [39]. The flat band potentials (E_{FB}) of $g-C_3N_4$ and NC- $g-C_3N_4$ -(0.01,

0.1, and 1.0) are -0.45, -0.49, -0.39 and -0.38 eV *vs* Ag/AgCl (-0.25, -0.29, -0.19 and -0.18 eV *vs* NHE). For n-type semiconductors, it is generally believed that the E_{CB} is about 0.1 eV lower than the E_{FB} value [40]. Therefore, the E_{CB} of g-C₃N₄ and NC-g-C₃N₄-(0.01, 0.1 and 1.0) can be approximately -0.35, -0.39, -0.29 and -0.28 eV *vs* NHE, respectively. Therefore, the E_{CB} of NC-g-C₃N₄-0.01 is negatively shifted by 0.04 eV compared to g-C₃N₄. Via the equation of $E_{VB} = E_g + E_{CB}$, and combining with the DRS band gap result of Fig. 7b, the valence band potentials (E_{VB}) of g-C₃N₄ and NC-g-C₃N₄-(0.01, 0.1 and 1.0) are estimated to be 2.14, 2.01 1.84 and 1.81 eV *vs* NHE, respectively (Fig. 8). Obviously, the band gap range of all of samples can satisfy the thermodynamic reduction potential of N₂ to NH₄⁺ (-0.092 eV *vs* NHE). Moreover, the potential of CB in NC-g-C₃N₄-0.01 is more negative than g-C₃N₄, NC-g-C₃N₄-0.1 and NC-g-C₃N₄-1.0, which facilitate photocatalytic nitrogen fixation to reduce N₂ to NH₄⁺.

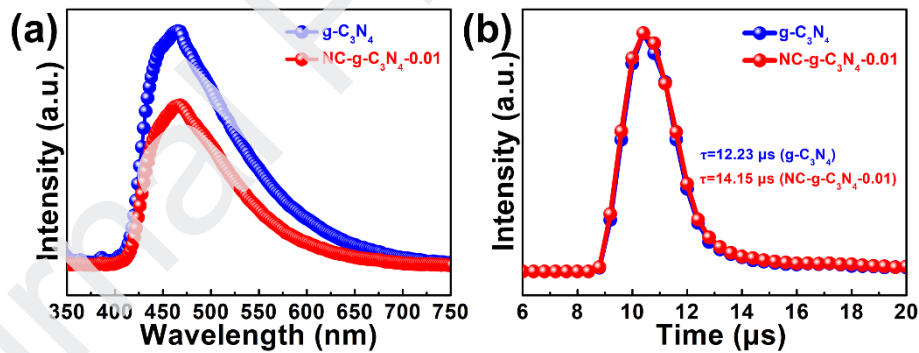


Fig. 9. (a) Steady photoluminescence (PL) and (b) time-resolved PL decay spectra of g-C₃N₄ and NC-g-C₃N₄-0.01 (λ_{ex} = 325 nm).

To know the function of photogenerated carriers, the PL spectra of NC-g-C₃N₄-0.01 were investigated [41]. The steady PL results are shown in Fig. 9a, wherein NC-g-C₃N₄-0.01 clearly exhibits much lower intensity than g-C₃N₄ [42], indicating that the recombination of carriers of NC-g-C₃N₄ can be substantially suppressed after

introduction of cyano groups. Since the cyano groups tend to trap photogenerated electrons, contributing to the separation of photoexcited carriers [43]. Besides, the exciton lifetime was calculated by fitting the time-resolved PL decay curves (Fig. 9b). The average intensity lifetimes (τ) of NC-g-C₃N₄-0.01 is 14.15 μ s, which is longer than g-C₃N₄ (12.23 μ s). This indicates that the increase of charge carrier lifetime in NC-g-C₃N₄-0.01 is in connection with the improvement of carrier separation. It is proved that adding KOH during the thermolysis facilitates the separation of photoexcited carriers in NC-g-C₃N₄. Consequently, the obtained NC-g-C₃N₄ exhibits longer photocarriers lifetime which increase the photocatalytic performance consequently.

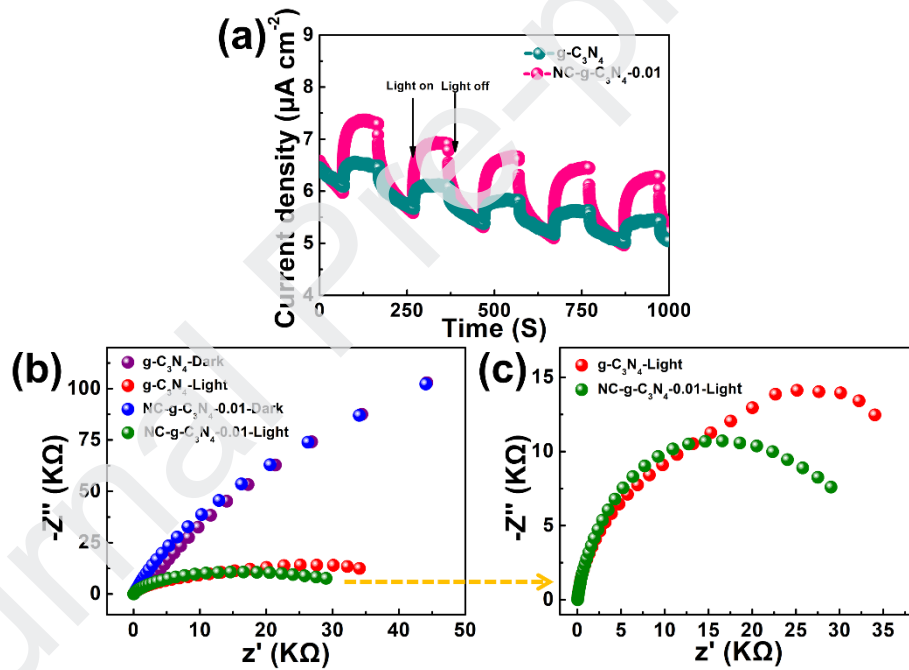


Fig. 10. (a) Photocurrent responses and (b, c) EIS of g-C₃N₄ and NC-g-C₃N₄-0.01.

In order to further study the separation of photoexcited carriers in NC-g-C₃N₄, transient photocurrent responses and EIS were performed [44]. As show in Fig. 10a, both g-C₃N₄ and NC-g-C₃N₄ display noticeable positive photocurrents by several on-off cycles [45]. Furthermore, the photocurrent intensity of NC-g-C₃N₄ is higher than g-

C_3N_4 , indicating NC-g- C_3N_4 has more efficient carrier separation by introducing cyano groups, which is in agreement with the PL measurements. In addition, the same results were obtained in the EIS measurements. Usually, the smaller radius of arc of the EIS spectrum curves indicates that the charge transfer resistance of electrode is smaller, and more efficient carrier separation is achieved [38]. Obviously, the impedance of samples in the dark is considerably greater than that under light illumination (Fig. 10b). And the arc sizes of the samples under illumination are NC-g- C_3N_4 < g- C_3N_4 (Fig. 10c), which imply higher carrier transfer efficiency of NC-g- C_3N_4 .

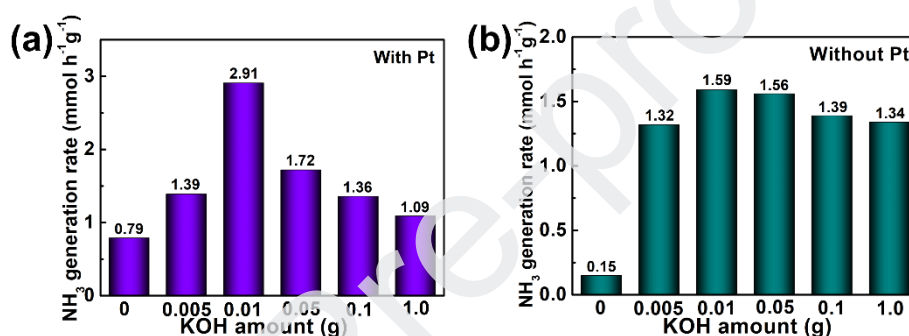
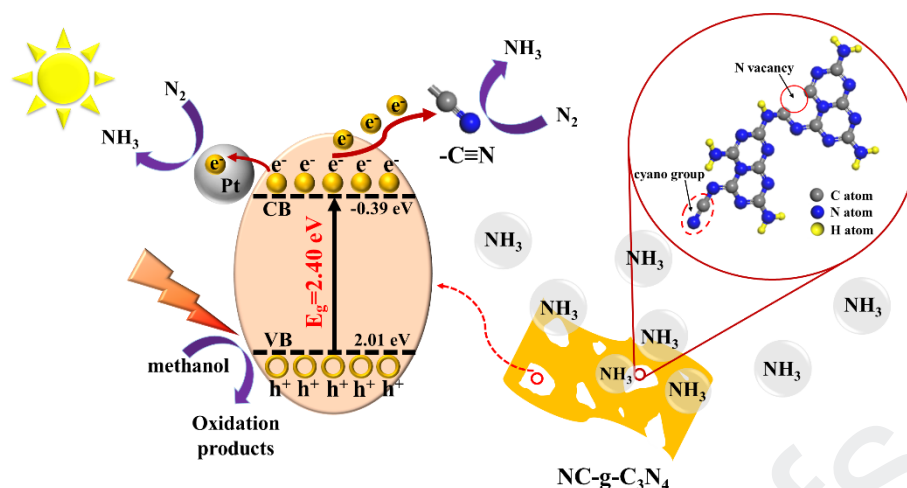


Fig. 11. The NH_3 generation rates of g- C_3N_4 and NC-g- C_3N_4 (a) with and (b) without Pt.

Herein, the photocatalytic N_2 fixation activity of NC-g- C_3N_4 is measured by the concentration of NH_4^+ . Fig. 11 show the NH_3 generation rate of the samples with Pt as co-catalysts and in the absence of Pt. In the N_2 atmosphere (Fig. 11a), NC-g- C_3N_4 -0.01 presents significantly higher photocatalytic nitrogen fixation activity ($2.91 \text{ mmol h}^{-1} \text{ g}^{-1}$) than pristine g- C_3N_4 ($0.79 \text{ mmol h}^{-1} \text{ g}^{-1}$) with Pt. The superior activities of NC-g- C_3N_4 is ascribed to its larger surface area, porous structure and nitrogen vacancies, which are responsible for capturing and activating N_2 . Besides, the cyano groups in NC-g- C_3N_4 can efficiently harvest photoexcited electrons and promote carrier separation.

Compared with other KOH usage samples, NC-g-C₃N₄-0.01 shows the best photocatalytic N₂ fixation activity (Fig. 11a), owing to its lower CB positions (Fig. 8b), causing decreased reduction driving force for N₂ fixation [45]. Interestingly, in the absence of Pt (Fig. 11b), the NC-g-C₃N₄-0.01 still have the excellent photocatalytic N₂ fixation activity (1.59 mmol h⁻¹g⁻¹), better than g-C₃N₄ and NC-g-C₃N₄ with other KOH usages. As shown in Table S3, by comparing with the other catalysts, our synthetic NC-g-C₃N₄-0.01 catalysts present the best photocatalytic N₂ fixation activity. In addition, we also carried out photocatalytic nitrogen fixation experiments with NC-g-C₃N₄ synthesized by other alkali metal compounds (NaOH and Ba(OH)₂). As shown in Fig. S8, the results show that there is no significant difference in nitrogen fixation rate between various alkali metal compounds (Ba(OH)₂, NaOH and KOH), which highlights the versatility of this preparation method.

To further investigate the nitrogen source of NH₄⁺, ¹⁵N isotopic labeling experiment is performed [46, 47]. As shown in Fig. S9, ¹H nuclear magnetic resonance spectra (¹H NMR) show a triplet coupling for ¹⁴NH₄⁺ and a doublet coupling for ¹⁵NH₄⁺ with ¹⁴N₂ and ¹⁵N₂ as the feeding gas, respectively. As a result, it is strongly confirmed that NH₃ is produced by NC-g-C₃N₄ photocatalytic fixation of N₂ instead of decomposition of g-C₃N₄.



Scheme 3. The mechanism of nitrogen photofixation of NC-g-C₃N₄.

As shown in Scheme 3, during photocatalytic nitrogen fixation, photoexcited electrons in the CB of NC-g-C₃N₄ transfer to the cyano groups and then reduce N₂ to NH₃. Meanwhile, the holes are consumed by the methanol. The presence of cyano groups, as electron acceptors, promotes the separation of photoexcited carriers. In this work, the narrower band gap upon nitrogen vacancies-introducing enable NC-g-C₃N₄ to utilize a larger part of visible light (Fig. 7a), thus more photons are absorbed, which is the main reason for improvement of the photocatalytic activity of NC-g-C₃N₄ [48-50]. Besides, the CB position for NC-g-C₃N₄ is more negative than pure g-C₃N₄ (Fig. 8b), indicating it is better able to N₂ fixation. The NC-g-C₃N₄ has higher specific surface area with abundant pores [51, 52], providing more active sites. Furthermore, the porous structure of NC-g-C₃N₄ can enhance light absorption by reflection/refraction of light. Thus, the porous structure, nitrogen vacancies, and cyano groups produce a synergistic effect, which induce effective carrier separation, longer carrier lifetime, more active sites and utilized larger part of visible light, endowing the improved activity of NC-g-C₃N₄.

3. Conclusions

In conclusion, porous structure, nitrogen vacancies and cyano groups are simultaneously introduced into g-C₃N₄ by the addition of alkali during the pyrolysis of urea, in which KOH promote the formation of porous structure, nitrogen vacancies and cyano groups. With the increase of KOH concentration from 0.005-0.01 g, the N₂ fixation activity is promoted. However, overlarge KOH concentrations (0.05, 0.1, 1.0 g) cause the destructive break and even decrease of N₂ fixation activity. The photocatalytic NH₃ generation rate of NC-g-C₃N₄ with an optimized KOH usage (0.01 g) with Pt as co-catalysts is nearly 3.7 times higher than g-C₃N₄. Interestingly, NC-g-C₃N₄ show significantly photocatalytic N₂ fixation activity (1.59 mmol h⁻¹g⁻¹) in the absence of co-catalysts. The excellent photocatalytic N₂ fixation performance of NC-g-C₃N₄ results from the synergistic effects, such as the improved light absorption and narrowed bandgap aroused by nitrogen vacancies, the promoted carrier transfer by cyano groups, and the adsorption of nitrogen on the exposed active sites by large BET surface area and porous structures.

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Graphical Abstract

